# Manage cluster mediators

ONTAP 9.9.1 REST API reference

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# Manage cluster mediators

## Cluster mediators endpoint overview

### Overview

You can use this API to add or remove a mediator to MetroCluster over IP configuration, or get the status and details of the existing mediator in MetroCluster over IP configuration. The GET operation returns the status of the mediator along with the mediator details. The DELETE operation removes the mediator. The POST operation adds the mediator.

### Adding a mediator

A mediator can be added to MetroCluster over IP configuration by issuing a POST on /cluster/mediators. Parameters are provided in the body of the POST request. There are no optional parameters for adding a mediator.

#### Required configuration fields

These fields are always required for any POST /cluster/mediators request.

- `ip_address` - Specifies the IP address of the mediator.
- `user` - Specifies a user name credential.
- `password` - Specifies a password credential.

#### Polling the setup job

After a successful POST /cluster/mediators is issued, an HTTP status code of 202 (Accepted) is returned along with a job UUID and a link in the body of the response. The setup job continues asynchronously and can be monitored by using the job UUID and the /cluster/jobs API. The "message" field in the response of the GET /cluster/jobs/{uuid} request shows the current step in the job, and the "state" field shows the overall state of the job.

### Deleting a Mediator

A mediator can be deleted from MetroCluster over IP configuration by issuing a DELETE to /cluster/mediators/{uuid}. Parameters are provided in the body of the DELETE request. There are no optional parameters for adding a mediator.

#### Required configuration fields

These fields are always required for any DELETE /cluster/mediators/{uuid} request.

- `user` - Specifies a user name credential.
- `password` - Specifies a password credential.

#### Polling the delete job

After a successful DELETE /cluster/mediators/{uuid} is issued, an HTTP status code of 202 (Accepted) is returned along with a job UUID and a link in the body of the response. The delete job continues

asynchronously and can be monitored by using the job UUID and the /cluster/jobs API. The "message" field in the response of the GET /cluster/jobs/{uuid} request shows the current step in the job, and the "state" field shows the overall state of the job.

## Examples

### Setting up a mediator for a 4-Node MetroCluster over IP Configuration

This example shows the POST body when setting up a mediator for a 4-Node MetroCluster over IP configuration. The only prerequisite is that MetroCluster over IP is configured.

```
# API
/api/cluster/mediators
```

### POST body included from file

```
mediator_post_body.txt:
{
"ip_address": "1.1.1.1",
"user": "username",
"password": "password"
}
curl -X POST https://<mgmt-ip>/api/cluster/mediators -d
"@mediator_post_body.txt"
```

### Inline POST body

```
curl -X POST https://<mgmt-ip>/api/cluster/mediators -H "Content-Type:
application/hal+json" -d '{"ip_address": "1.1.1.1", "user": "username",
"password": "password"}'
```

### POST Response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 07:40:59 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 189
Content-Type: application/hal+json
{
"job": {
  "uuid": "f567b48b-fca6-11ea-acaf-005056bb47c1",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/f567b48b-fca6-11ea-acaf-005056bb47c1"
    }
  }
}
}
```

### Monitoring the job progress

Use the link provided in the response to the POST request to fetch information for the mediator setup job.

#### Request

```
curl -X GET https://<mgmt-ip>/api/cluster/jobs/f567b48b-fca6-11ea-acaf-
005056bb47c1
```

#### Job status response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 07:41:29 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 189
Content-Type: application/hal+json
{
"uuid": "f567b48b-fca6-11ea-acaf-005056bb47c1",
"description": "POST /api/cluster/mediators/",
"state": "running",
"start_time": "2020-09-22T03:41:00-04:00",
"_links": {
  "self": {
    "href": "/api/cluster/jobs/f567b48b-fca6-11ea-acaf-005056bb47c1"
  }
}
}
```

**Final status of a successful Mediator add**

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 07:43:38 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 358
Content-Type: application/hal+json
{
"uuid": "f567b48b-fca6-11ea-acaf-005056bb47c1",
"description": "POST /api/cluster/mediators/",
"state": "success",
"message": "success",
"code": 0,
"start_time": "2020-09-22T03:41:00-04:00",
"end_time": "2020-09-22T03:42:10-04:00",
"_links": {
  "self": {
    "href": "/api/cluster/jobs/f567b48b-fca6-11ea-acaf-005056bb47c1"
  }
}
}
```

### Retrieving the existing mediator configurations

#### Request

```
curl -X GET https://<mgmt-ip>/api/cluster/mediators
```

#### Response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 08:53:18 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 320
Content-Type: application/hal+json
{
"records": [
  {
    "uuid": "f89e8906-fca6-11ea-acaf-005056bb47c1",
    "_links": {
      "self": {
        "href": "/api/cluster/mediators/f89e8906-fca6-11ea-acaf-
005056bb47c1"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/cluster/mediators"
  }
}
}
```

### Retrieving a specific mediator using the uuid

#### Request

```
curl -X GET https://<mgmt-ip>/api/cluster/mediators/f89e8906-fca6-11ea-
acaf-005056bb47c1
```

#### Response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 08:59:40 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 347
Content-Type: application/hal+json
{
"uuid": "f89e8906-fca6-11ea-acaf-005056bb47c1",
"ip_address": "10.234.173.40",
"port": 31784,
"reachable": true,
"peer_cluster": {
  "name": "mcc_siteB",
  "uuid": "38779fd1-fc6b-11ea-9421-005056bb21d8"
},
"_links": {
  "self": {
    "href": "/api/cluster/mediators/f89e8906-fca6-11ea-acaf-005056bb47c1"
  }
}
}
```

### Deleting a configured Mediator using the uuid

#### Request

```
curl -X DELETE https://<mgmt-ip>/api/cluster/mediators/{uuid} -H "Content-
Type: application+hal/json" -d '{"user": "username", "password":
"password"}'
```

#### Response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 09:13:52 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 189
Content-Type: application/hal+json
{
"job": {
  "uuid": "eeb71ccd-fcb3-11ea-acaf-005056bb47c1",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/eeb71ccd-fcb3-11ea-acaf-005056bb47c1"
    }
  }
}
}
```

### Monitoring the job progress

Use the link provided in the response to the DELETE request to fetch information for the delete job.

#### Request

```
curl -X GET https://<mgmt-ip>/api/cluster/jobs/eeb71ccd-fcb3-11ea-acaf-
005056bb47c1
```

#### Job status response

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 09:14:20 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 316
Content-Type: application/hal+json
{
"uuid": "eeb71ccd-fcb3-11ea-acaf-005056bb47c1",
"description": "DELETE /api/cluster/mediators/f89e8906-fca6-11ea-acaf-
005056bb47c1",
"state": "running",
"start_time": "2020-09-22T05:13:52-04:00",
"_links": {
  "self": {
    "href": "/api/cluster/jobs/eeb71ccd-fcb3-11ea-acaf-005056bb47c1"
  }
}
}
```

**Final status of the Mediator DELETE job**

```
HTTP/1.1 202 Accepted
Date: Tue, 22 Sep 2020 09:21:46 GMT
Server: libzapid-httpd
X-Content-Type-Options: nosniff
Cache-Control: no-cache,no-store,must-revalidate
Location: /api/cluster/metrocluster
Content-Length: 396
Content-Type: application/hal+json
{
"uuid": "eeb71ccd-fcb3-11ea-acaf-005056bb47c1",
"description": "DELETE /api/cluster/mediators/f89e8906-fca6-11ea-acaf-
005056bb47c1",
"state": "success",
"message": "success",
"code": 0,
"start_time": "2020-09-22T05:13:52-04:00",
"end_time": "2020-09-22T05:14:24-04:00",
"_links": {
  "self": {
    "href": "/api/cluster/jobs/eeb71ccd-fcb3-11ea-acaf-005056bb47c1"
  }
}
}
```

# Retrieve ONTAP Mediators configured in the cluster

GET `/cluster/mediators`

**Introduced In:** 9.8

Retrieves mediators configured in the cluster.

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| peer_cluster.uuid | string | query | False | Filter by peer_cluster.uuid |
| peer_cluster.name | string | query | False | Filter by peer_cluster.name |
| reachable | boolean | query | False | Filter by reachable |
| uuid | string | query | False | Filter by uuid |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| port | integer | query | False | Filter by port |
| ip_address | string | query | False | Filter by ip_address |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | |
| records | array[records] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "dr_group": {
      "id": 0
    },
    "ip_address": "10.10.10.7",
    "password": "mypassword",
    "peer_cluster": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "cluster2",
      "uuid": "ebe27c49-1adf-4496-8335-ab862aebebf2"
    },
    "port": "31784",
    "reachable": 1,
    "user": "myusername",
    "uuid": "string"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

dr_group

DR group reference.

| Name | Type | Description |
|---|---|---|
| id | integer | DR Group ID |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

peer_cluster

The peer cluster that the mediator service is used for.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

records

| Name | Type | Description |
|---|---|---|
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true |
| dr_group | dr_group | DR group reference. |

| Name | Type | Description |
|---|---|---|
| ip_address | string | The IP address of the mediator. |
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create and connect an ONTAP Mediator

POST `/cluster/mediators`

**Introduced In:** 9.8

Creates and connect a mediator.

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| return_records | boolean | query | False | The default is false. If set to true, the records are returned.<br>• Default value: |

## Request Body

| Name | Type | Description |
| --- | --- | --- |
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true<br>• Introduced in: 9.8 |
| dr_group | dr_group | DR group reference. |
| ip_address | string | The IP address of the mediator. |
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

**Example request**

```
{
  "dr_group": {
    "id": 0
  },
  "ip_address": "10.10.10.7",
  "password": "mypassword",
  "peer_cluster": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "cluster2",
    "uuid": "ebe27c49-1adf-4496-8335-ab862aebebf2"
  },
  "port": "31784",
  "reachable": 1,
  "user": "myusername",
  "uuid": "string"
}
```

## Response

```
Status: 202, Accepted
```

| Name | Type | Description |
|---|---|---|
| job | job_link | |

#### Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

## Error

```
Status: Default
```

ONTAP Error Response codes

| Error code | Description |
|---|---|
| 13369351 | Update to mediator failed. Reason: does not authorized for that command. Check that the peer cluster and mediator are reachable. |

| Name | Type | Description |
|---|---|---|
| error | error | |

#### Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

dr_group

DR group reference.

| Name | Type | Description |
|---|---|---|
| id | integer | DR Group ID |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

peer_cluster

The peer cluster that the mediator service is used for.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

mediator

Mediator information

| Name | Type | Description |
|---|---|---|
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true<br>• Introduced in: 9.8 |
| dr_group | dr_group | DR group reference. |
| ip_address | string | The IP address of the mediator. |

| Name | Type | Description |
| --- | --- | --- |
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

job_link

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| uuid | string | The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |

| Name | Type | Description |
| --- | --- | --- |
| target | string | The target parameter that caused the error. |

# Delete an ONTAP Mediator

DELETE `/cluster/mediators/{uuid}`

**Introduced In:** 9.8

Deletes the mediator.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| uuid | string | path | True | • format: uuid |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |

## Request Body

| Name | Type | Description |
|---|---|---|
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true<br>• Introduced in: 9.8 |
| dr_group | dr_group | DR group reference. |
| ip_address | string | The IP address of the mediator. |

| Name | Type | Description |
|---|---|---|
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

**Example request**

```
{
  "dr_group": {
    "id": 0
  },
  "ip_address": "10.10.10.7",
  "password": "mypassword",
  "peer_cluster": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "cluster2",
    "uuid": "ebe27c49-1adf-4496-8335-ab862aebebf2"
  },
  "port": "31784",
  "reachable": 1,
  "user": "myusername",
  "uuid": "string"
}
```

## Response

```
Status: 202, Accepted
```

| Name | Type | Description |
|---|---|---|
| job | job_link | |

### Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

## Error

```
Status: Default
```

ONTAP Error Response codes

| Error code | Description |
|---|---|
| 13369377 | Mediator field "mediator.id" does not exist. |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

dr_group

DR group reference.

| Name | Type | Description |
|---|---|---|
| id | integer | DR Group ID |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

peer_cluster

The peer cluster that the mediator service is used for.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

mediator

Mediator information

| Name | Type | Description |
|---|---|---|
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true<br>• Introduced in: 9.8 |
| dr_group | dr_group | DR group reference. |
| ip_address | string | The IP address of the mediator. |

| Name | Type | Description |
|---|---|---|
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

job_link

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |

| Name | Type | Description |
|---|---|---|
| target | string | The target parameter that caused the error. |

# Retrieve the ONTAP Mediator state and configuration

GET `/cluster/mediators/{uuid}`

**Introduced In:** 9.8

Retrieves the mediator state and configuration.

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | • format: uuid |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| ca_certificate | string | CA certificate for ONTAP Mediator. This is optional if the certificate is already installed.<br>• x-ntap-createOnly: true<br>• Introduced in: 9.8 |
| dr_group | dr_group | DR group reference. |
| ip_address | string | The IP address of the mediator. |
| password | string | The password used to connect to the REST server on the mediator. |
| peer_cluster | peer_cluster | The peer cluster that the mediator service is used for. |

| Name | Type | Description |
|---|---|---|
| port | integer | The REST server's port number on the mediator. |
| reachable | boolean | Indicates the connectivity status of the mediator. |
| user | string | The username used to connect to the REST server on the mediator. |
| uuid | string | The unique identifier for the mediator service. |

**Example response**

```
{
  "dr_group": {
    "id": 0
  },
  "ip_address": "10.10.10.7",
  "password": "mypassword",
  "peer_cluster": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "cluster2",
    "uuid": "ebe27c49-1adf-4496-8335-ab862aebebf2"
  },
  "port": "31784",
  "reachable": 1,
  "user": "myusername",
  "uuid": "string"
}
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

dr_group

DR group reference.

| Name | Type | Description |
|---|---|---|
| id | integer | DR Group ID |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

peer_cluster

The peer cluster that the mediator service is used for.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |

| Name | Type | Description |
| --- | --- | --- |
| target | string | The target parameter that caused the error. |

# Retrieve historical performance metrics for the cluster

GET `/cluster/metrics`

**Introduced In:** 9.6

Retrieves historical performance metrics for the cluster.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| timestamp | string | query | False | Filter by timestamp |
| throughput.total | integer | query | False | Filter by throughput.total |
| throughput.other | integer | query | False | Filter by throughput.other |
| throughput.read | integer | query | False | Filter by throughput.read |
| throughput.write | integer | query | False | Filter by throughput.write |
| latency.total | integer | query | False | Filter by latency.total |
| latency.other | integer | query | False | Filter by latency.other |
| latency.read | integer | query | False | Filter by latency.read |
| latency.write | integer | query | False | Filter by latency.write |
| duration | string | query | False | Filter by duration |
| iops.total | integer | query | False | Filter by iops.total |
| iops.other | integer | query | False | Filter by iops.other |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| iops.read | integer | query | False | Filter by iops.read |
| iops.write | integer | query | False | Filter by iops.write |
| status | string | query | False | Filter by status |
| interval | string | query | False | The time range for the data. Examples can be 1h, 1d, 1m, 1w, or 1y. The period for each time range is specified as follows:<br>• 1h: Metrics over the most recent hour sampled over 15 seconds.<br>• 1d: Metrics over the most recent day sampled over 5 minutes.<br>• 1w: Metrics over the most recent week sampled over 30 minutes.<br>• 1m: Metrics over the most recent month sampled over 2 hours.<br>• 1y: Metrics over the most recent year sampled over a day.<br>• Default value: 1<br>• enum: ["1h", "1d", "1w", "1m", "1y"] |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |
| desc] direction. Default direction is 'asc' for ascending. | return_records | boolean | query | False |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[records] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "duration": "PT15S",
    "iops": {
      "read": "200",
      "total": "1000",
      "write": "100"
    },
    "latency": {
      "read": "200",
      "total": "1000",
      "write": "100"
    },
    "status": "ok",
    "throughput": {
      "read": "200",
      "total": "1000",
      "write": "100"
    },
    "timestamp": "2017-01-25T11:20:13Z"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

iops

The rate of I/O operations observed at the storage object.

| Name | Type | Description |
|---|---|---|
| other | integer | Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on. |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

latency

The round trip latency in microseconds observed at the storage object.

| Name | Type | Description |
| --- | --- | --- |
| other | integer | Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on. |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

throughput

The rate of throughput bytes per second observed at the storage object.

| Name | Type | Description |
| --- | --- | --- |
| other | integer | Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on. |
| read | integer | Performance metric for read I/O operations. |
| total | integer | Performance metric aggregated over all types of I/O operations. |
| write | integer | Peformance metric for write I/O operations. |

records

Performance numbers, such as IOPS latency and throughput.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| duration | string | The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations: |
| iops | iops | The rate of I/O operations observed at the storage object. |
| latency | latency | The round trip latency in microseconds observed at the storage object. |
| status | string | Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data. |
| throughput | throughput | The rate of throughput bytes per second observed at the storage object. |
| timestamp | string | The timestamp of the performance data. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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